

1. Please refer to the development process of Intel microprocessors, and try to speculate that with the advancement of integrated circuit process technology, processors consume less and less power, but why do most processors need to use cooling fans? (10%)

Sol: As the number of transistors grows rapidly in compliance with Moore's Law, the power consumed by a large number of transistors in the microprocessor also increases rapidly, so cooling fans are needed to assist in heat dissipation.

2. What is a command set computer? How does such a concept help or benefit computer design? (10%)

sol: instruction set computer:

An electronic computer that integrates all hardware operating circuits is called an instruction set computer.

It is easy to control the computer with programmed hardware circuits, making it more flexible to use.

**3. Please explain the following terms: (1) Response time
(2) Execution time (3) Throughput (4) Benchmark(10%)**

Sol:

- (1) **Response time**: How long does it take for the first response to be generated when the required work is submitted.
- (2) **Execution time**: The time it takes for a job to run from start to finish.
- (3) **Generate**: The maximum number of times a specific task can be completed per unit of time.
- (4) **Evaluation program**: A set of hybrid programs that cover various calculations and are used to evaluate system performance is called a set of evaluation programs. The single program that makes up the review program group is called the evaluation program.

4. The performance of a P microprocessor with 100 MHz was evaluated by executing 10,000,000 evaluation program instructions, and it was found that it took 0.25 seconds. What are the CPI and MIPS data during this performance experiment? (10%)

Sol:

$$(1) \text{ CPI} = \frac{0.25 \times 100 \times 10^6}{10 \times 10^6} = 2.5$$

$$(2) \text{ MIPS} = \frac{10 \times 10^6}{0.25 \times 10^6} = 40$$

5. There is a 500MHz processor, and the following is the data obtained after measurement

Instruction set	CPI	frequency of use
A	2	40%
B	3	25%
C	3	25%
D	3	10%

(1) **What**
is the MIPS of this processor, please? (10%)

(2) **What**
is the CPI of this processor, please? (10%)

Sol:

$$(1) \quad \text{CPI} = 2 \times 0.4 + 3 \times 0.25 + 3 \times 0.25 + 3 \times 0.1 = 2.6$$

$$\text{MIPS} = \frac{500 \times 10^6}{2.6 \times 10^6} = 192.307$$

$$(2) \quad \text{CPI} = 2.6$$

**6. Please try to explain the program compilation period
(compile time)What are the main tasks included?
Why does a program have to go through this stage
before it can be executed? (10%)**

Sol: Main work:

- Convert high-level language programs to low-level language programs for specific computers.

Non-enforceable reasons:

- Computers cannot run high-level languages, so they must be compiled into machine code that can be executed by the computer through compilation time.

7. Using the stacked structure, the diagram illustrates the process of generating

post-fix representations in the following formulas

(a) and pre-fix representations in the following formulas (b)

(a) $A / B - C + D * E + F * G$ (15%)

(b) $A * (B + C) - D$ (15%)

Ans: (a) **post-fix : $A B / C - D E * + F G * +$**

(b) **Pre-fix : $- * A + B C D$**